

MAT 157: Week 16

Summary. This week is dedicated to the **Fundamental Theorem of Calculus (FTC)**.

Here it is, with precise assumptions:

- (I) (*differentiating the integral.*) If f is integrable on $[a, b]$ and continuous at some point $c \in (a, b)$, then $F(x) := \int_a^x f(t) dt$ is differentiable at c and $F'(c) = f(c)$. At the end-points we have the one-sided derivatives $F'(a^+) = f(a)$ and $F'(b^-) = f(b)$.
- (II) (*integrating the derivative.*) If F is continuous on $[a, b]$, differentiable on (a, b) , and $f := F'$ is integrable on $[a, b]$, then $\int_a^b f(t) dt = F(b) - F(a)$.

In many applications, the following will do:

(I/II) (*convenient version.*) Let $F, f : [a, b] \rightarrow \mathbb{R}$. If f is **continuous** on $[a, b]$ then

$$\left. \begin{array}{l} F' = f \text{ on } (a, b), \\ F'(a^+) = f(a), \text{ and } F'(b^-) = f(b) \end{array} \right\} \iff F(x) = \int_a^x f(t) dt + C \text{ on } [a, b]$$

for some $C \in \mathbb{R}$. The lower limit of integration can be changed to any $c \in [a, b]$ (which would then change the constant to $C + \int_a^c f$).

Recalling that $\int f$ denotes the set of anti-derivatives (or primitives) of f , if F is an antiderivative of f then $\int f = \{F(x) + C \mid C \in \mathbb{R}\}$. In general, one cannot expect to be able to find the antiderivatives of a function. However, in Chapter 19 you learnt about methods that can apply in some situations: substitution, integration by parts, and partial fractions. As well as guessing.

Substitution: Certainly $\int (f \circ g)' = f \circ g + C$ for some C and so $\int f'(g(x))g'(x) dx = f(g(x)) + C$. Given an integral, you may be able to express the integrand as the product of two functions, $f'(g(x))$ and $g'(x)$, and then you're done!

Integration by Parts: Certainly $\int fg + C = \int (fg)' = \int f'g + \int fg'$ and so $\int fg' = fg - \int f'g$. Given an integral, you may be able to express the integrand as the product of two functions, f and g' , one of which is easy to integrate and the other of which differentiates in a useful manner, and then you can move forward!

Partial Fractions: If you're trying to find an antiderivative of a rational function, first do long division so that you can express it as a polynomial plus a rational function whose numerator is lower degree than its denominator. Factor the denominator over the reals, so that it's the product of terms like $(x-5)$, $(x+2)^3$, $((x+5)^2+6)$, and $((x-2)^2+4)^4$. Express the rational function as a linear combination of rational functions whose denominators have only one term of that form. Integrate.

Note that we did not discuss the conditions on f and g that would make substitution and integration by parts valid. We encourage you to apply the methods in a formal manner (i.e. don't worry about validity) and if you do end up with an expression you think to be an antiderivative, check that it works. If it does, you can proceed — you don't need to defend the validity of how you found it.

Read: Chapters 14 and 19.

For further practice: Chapter 14 Problems 1 and 2 (a few parts each), 5 (a), 6, 7 (i), 13, 19. Chapter 19 Problems 1-3 (a few parts each), 5, 6, 15, 21, 24, 26. Note: we do not expect you to be able to integrate things that involve inverse trig functions (yet) and so some of the partial fraction problems and problems like the one on the second half of page 375 are currently out of reach.

Hand-in (due Friday, February 13 by 9am):

1. (see Spivak Problem 14.1). Find the derivatives of the following functions.

(a) $F(x) = \sin \left(\int_0^x \sin \left(\int_0^y \sin^3(t) dt \right) dy \right)$ (c) $G(x) = \int_x^b \frac{1}{1+t^2 + \sin^2 t} dt$

(b) $H(x) = \int_2^x \left(\int_0^y G(t) dt \right) dy$ (d) $K(x) = \int_3^{\int_0^x \sin^3 t dt} \frac{1}{1+t^2} dt.$

2. *Some Integration Methods.* Find the following indefinite integrals and verify they're correct.

(a) $\int \frac{2x^2 + x + 1}{(x+3)(x-1)^2} dx$ (c) $\int \frac{\log(\log(x))}{x \log(x)} dx$

(b) $\int e^{ax} \sin bx dx$ (d) $\int (\log(x))^3 dx.$

3. *Mean Value Theorem of Integration.* Given a continuous function $f : [a, b] \rightarrow \mathbb{R}$, let

$$\bar{f} := \frac{1}{b-a} \int_a^b f(x) dx \quad \text{denote its average over } [a, b].$$

- (a) Prove that there exists a point $c \in [a, b]$ such that $f(c) = \bar{f}$.
(b) Some students are taught that $\frac{x(t_2) - x(t_1)}{t_2 - t_1}$ is the “average rate of change” of $x(t)$ over the interval $[t_1, t_2]$. Justify this naming, making sure to specify needed conditions on x .

4. I is an interval in $\mathbb{R}$, f is continuous on I , and $a \in I$. Let $G(x) := \int_a^x (x-t)f(t) dt$.

- (a) Find $G'(x)$ and prove $G(x) = \int_a^x \int_a^u f(t) dt du$. Now prove it using integration by parts.
(b) Find $G'''(x)$.
(c) You seek an expression for a function whose n th derivative is f . But you can only afford one integral sign at the Calculus Store. What formula will you build? (No proof needed.)

5. *Long-term integrals*

- (a) Suppose that f is integrable on $[0, x]$ for every $x > 0$, and that $\lim_{x \rightarrow \infty} f(x) = a$. Prove that this limit also equals the long-term average:

$$\lim_{x \rightarrow \infty} \frac{1}{x} \int_0^x f(t) dt = a.$$

Hint: For $N > 0$ (to be chosen later) and $x > N$, write $x = N + y$ and split the integral.

- (b) Prove that $\int_0^x \frac{\sin t}{1+t} dt > 0$ for all $x > 0$ and that $\lim_{x \rightarrow \infty} \int_0^x \frac{\sin t}{1+t} dt$ exists.

6. *The Fundamental Lemma in the Calculus of Variations*

Suppose that f is continuous on $[a, b]$, and $\int_c^d f = 0$ for every subinterval $[c, d] \subset [a, b]$.

Prove that $f(x) = 0$ for all $x \in [a, b]$, in two ways:

- (a) using the definition of the integral (by upper and lower sums);
(b) using the Fundamental Theorem of Calculus.